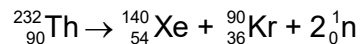


30 Thorium-232 undergoes the following fission reaction:



The binding energy per nucleon for the nuclei involved are:

Thorium-232: 7.6 MeV

Xenon-140: 8.3 MeV

Krypton-90: 8.5 MeV

What is the energy released by this fission reaction?

A	$1.64 \times 10^8 \text{ J}$	B	$2.63 \times 10^{-11} \text{ J}$	C	$1.47 \times 10^{-12} \text{ J}$	D	$2.63 \times 10^{-17} \text{ J}$
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Answer: B

Explanation/Working:

Thorium-232 has 232 nucleons.

Binding energy = $232 \times 7.6 = 1763.2 \text{ MeV}$

Xenon-140: $140 \times 8.3 = 1162.0 \text{ MeV}$

Krypton-90: $90 \times 8.5 = 765.0 \text{ MeV}$

$\begin{aligned}\text{Energy released} &= \text{Binding energy of products} - \text{Binding energy of reactants} \\ &= 1927.0 - 1763.2 = 163.8 \text{ MeV}\end{aligned}$
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$$1 \text{ MeV} = 1.602 \times 10^{-13} \text{ J}$$

$$\text{Energy in joules} = 163.8 \times 1.602 \times 10^{-13} = 2.63 \times 10^{-11} \text{ J}$$

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